

Clinical Study Summary Report (CSSR)

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1. Administrative & Study Identification

Official Study Title: Palbociclib Treatment Patterns and Outcomes in HR+/HER2- MBC: Flatiron Database Analysis **Brief Study Title:** A Real-life Study to Understand the Use and Effects of Palbociclib in US Patients With Breast Cancer **Protocol Number:** A5481189 **NCT Number:** NCT06495164 **Posting ID:** 6258097189184952 **Sponsor/Company Name:** Pfizer **Trial Status:** RECRUITING **Investigational Product:** Palbociclib (Ibrance), Ribociclib (Kisqali), Abemaciclib (Verzenio) **Phase of Development:** Not Applicable (Retrospective Observational Cohort Study) **Medical Monitor:** N/A (Retrospective Database Analysis)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate and compare the real-world effectiveness (Overall Survival) of palbociclib, ribociclib, and abemaciclib in combination with an aromatase inhibitor (AI) as first-line treatments for patients with HR+/HER2- metastatic breast cancer (MBC). **Secondary Objectives:** To understand how palbociclib combination therapy is used in real-life conditions, determine treatment duration for different CDK4/6 inhibitors, assess the occurrence of adverse events (such as neutropenia, anemia, and thrombocytopenia), and evaluate subsequent therapy patterns (e.g., use of chemotherapy later).

2.2 Background & Rationale Palbociclib, the first oral CDK4/6 inhibitor, is an approved medicine indicated for the treatment of a kind of advanced/metastatic breast cancer (MBC), called hormone receptors positive (HR+)/ Human epidermal growth factor receptor 2 negative (HER2-) disease. Palbociclib is given orally in combination with hormonal therapies. While progression-free survival (PFS) benefits have been robustly demonstrated for CDK4/6 inhibitors in phase 3 trials, overall survival (OS) data have been mixed. This study utilizes the US nationwide Flatiron Health electronic health record database to address a critical gap by providing real-world evidence (RWE) on comparative effectiveness, long-term survival, and utilization patterns among the three commercially available CDK4/6 inhibitors in routine clinical practice.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational (Retrospective cohort study using electronic health records) **Control Type:** Active Comparator (Palbociclib vs. Ribociclib vs. Abemaciclib)

Blinding: Open-label / Unblinded **Randomization:** N/A (Stabilized inverse probability of treatment weighting [sIPTW] used to adjust for confounding variables)

3.2 Planned Enrollment & Duration Targeted Enrollment (\$N\$): 5087 eligible patients **Number of Sites:** Over 280 cancer clinics (~800 sites of care) across the US **Estimated Study Start:** 24-Jun-2024 **Estimated Primary Completion:** 30-Dec-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male and female patients aged ≥ 18 years of age or older at the time of MBC diagnosis. 2. Confirmed diagnosis of hormone receptor-positive (HR+) / human epidermal growth factor receptor 2-negative (HER2-) metastatic breast cancer. 3. Initiated first line therapy (CDK4/6i, ET, CT, or other) in the metastatic setting during the period from February 2015 through June 2022 or data cutoff.

4.2 Exclusion Criteria 1. Patients without confirmed HR+/HER2-status. 2. Patients who received index treatment outside the specified data coverage period. 3. Missing essential baseline demographic or clinical data precluding sIPTW adjustment.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Dosed according to routine clinical oncology practice. **Route of Administration:** Oral (CDK4/6 inhibitors + Aromatase Inhibitors). **Duration of Treatment:** Variable; defined as the time from the start to the end of the index therapy or data cutoff/death.

5.2 Concomitant & Prohibited Therapy Permitted: Hormonal therapy, aromatase inhibitors (e.g., letrozole, anastrozole, exemestane) or other standard-of-care endocrine therapies. **Prohibited:** N/A (Real-world observational setting).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Index Date	Baseline	First-line therapy initiation; extraction of demographic and clinical data
Follow-up	Longitudinal	Retrospective EHR data tracking for OS, rwPFS, AEs, and subsequent therapies
End of Study	Data Cutoff	Followed until data cutoff, death, or last medical activity

(Note: As this is a retrospective database analysis, no physical patient visits are conducted).

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes 1. **Overall Survival (OS)**, defined as the time from index treatment initiation to death from any cause. Measurement Method: Kaplan-Meier estimates and Cox regression models utilizing electronic health records.

7.2 Secondary & Exploratory Outcomes 1. **Real Progression Free Survival (rwPFS)** 2. **Progression Free Survival** 2 3. **Real-World Tumor Response (rwTR)** 4. Treatment duration (Time to Treatment Discontinuation). 5. Incidence of adverse events (neutropenia, anemia, thrombocytopenia). 6. Subsequent treatment patterns (e.g., time to chemotherapy).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Extracted retrospectively from unstructured clinical notes and structured EHR data. **Serious Adverse Events (SAEs):** Evaluated based on recorded incidence of severe cytopenias (e.g., neutropenia, anemia). **Data Monitoring Committee (DMC):** N/A (Retrospective observational study).

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: All eligible adult patients in the Flatiron Health database initiating 1L therapy for HR+/HER2- MBC. **Sample Size Justification:** A large real-world cohort (N = 5087) was utilized to provide sufficient power to detect differences in OS across active treatment groups. **Handling of Missing Data:** Stabilized inverse probability of treatment weighting (sIPTW) applied to balance baseline demographics and clinical characteristics across cohorts.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: Conducted in accordance with standard observational research practices and HIPAA regulations regarding de-identified patient data. **Monitoring Plan:** Centralized database quality checks and curation by Flatiron Health. **Audit Trail:** Longitudinal tracking of EHR data processing, abstraction algorithms, and derivation mapping.

11. Study Identifiers & Governance

Primary ID: A5481189 **Posting ID:** 6258097189184952 **Registry Number:** NCT06495164 **Sponsor/Lead Organization:** Pfizer **Brief Title:** A Real-life Study to Understand the Use and Effects of Palbociclib in US Patients With Breast Cancer **Official Regulatory**

12. Clinical Framework (Breast Cancer Specific)

Disease Areas: Breast Cancer, Malignant Neoplasm of Breast
Primary Condition: HR+/HER2- Metastatic Breast Cancer (MBC)
Diagnosis Threshold: Histologically/cytologically confirmed HR+/HER2- MBC
Medical Methodology: First-line CDK4/6 inhibitor + Aromatase inhibitor therapy or related hormonal therapy
Optimization Target: Overall Survival and Treatment Duration

13. Study Procedures & Methodology

Management Pathway: Standard clinical oncology pathway for MBC
Education Protocol (HCP): N/A (Retrospective database study)
Education Protocol (Patient): N/A (Retrospective database study)
Quality Audit Mechanism: Flatiron Health EHR data curation and validation

14. Observation & Follow-Up Schedule

Total Duration: Retrospective follow-up until data cutoff, death, or last medical activity
Onsite Visit Cadence: N/A (Real-World Data)
Remote Monitoring Frequency: N/A (Real-World Data)
Checklist Review Interval: N/A (Real-World Data)

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				